



Internship Roadmap (AWT)

Domain: Python • Streamlit • Intro to LLMs (Gemini – basic usage)

Duration: 6–8 Weeks

Level: Beginner → Practical

Outcome: 2–3 mini projects + 1 capstone app



Week 1: Python Fundamentals (Foundation)



Objective

Build strong Python basics required for data handling and app logic.



Topics Covered

- Python installation & environment setup
- Variables, data types (int, float, string, list, dict, tuple)
- Conditional statements (if / else)
- Loops (for, while)
- Functions & basic modules
- Exception handling (try / except)
- File handling (read/write `.txt`, `.csv`)



Practice Tasks

- Calculator program
 - Student marks management script
 - Read/write data from a file
-



Week 2: Python for Data Handling



Objective

Learn how Python works with real-world data.



Topics Covered

- Introduction to `pip`
- Using libraries:
 - `pandas` (dataframes, CSV handling)

- `datetime`
- *Basic data cleaning*
- *Simple logic-based data analysis*

Mini Task

- *Read CSV → clean data → show summary*
 - *Simple report generation using Python*
-



Week 3: Streamlit Basics (UI + Logic)

Objective

Convert Python scripts into interactive web apps.

Topics Covered

- *What is Streamlit & use cases*
- *Streamlit app structure*
- *UI components:*
 - `st.text_input`
 - `st.button`
 - `st.selectbox`
 - `st.file_uploader`
- *Displaying data:*
 - `st.write`
 - `st.table`
 - `st.dataframe`
- *App state basics*

Mini Project 1

CSV Analyzer App

- *Upload CSV*
 - *Show data preview*
 - *Display basic stats*
-



Week 4: Streamlit + Business Logic

Objective

Build slightly complex apps using Python + Streamlit.

Topics Covered

- **Multi-page Streamlit apps**
- **Form-based inputs**
- **Basic charts (line, bar)**
- **Error handling in UI**
- **Structuring code (services, utils)**

Mini Project 2

Student Result Dashboard

- **Input marks**
 - **Calculate grade**
 - **Display result summary**
 - **Download report**
-

Week 5: Introduction to LLMs (Gemini – Very Simple)

Objective

Understand what LLMs are and how to make basic API calls.

Concepts (Simple Explanation)

- **What is an LLM?**
- **What Gemini does (text generation, Q&A)**
- **Prompt → Response flow**
- **API key usage (environment variables)**
- **Ethical & usage limitations**

Technical Topics

- **Gemini API setup**
- **Simple text prompt**
- **Handling response safely**
- **Error handling**

Practice Tasks

- **Ask Gemini to:**
 - **Summarize text**

- *Answer basic questions*
 - *Rewrite content*
-



Week 6: Streamlit + Gemini Integration



Objective

Create AI-powered UI apps.



Topics Covered

- *Connecting Streamlit UI with Gemini API*
- *Input → prompt → response → display*
- *Loading states & errors*
- *Prompt hygiene (simple rules)*



Mini Project 3



AI Text Assistant

- *User enters text*
 - *Selects task:*
 - *Summarize*
 - *Rewrite*
 - *Explain*
 - *Gemini generates output*
-



Week 7–8: Final Capstone Project



Objective

Apply everything learned in one complete project.



Project Options (Choose One)

1. *AI Resume Helper*
2. *AI Notes Summarizer*
3. *AI Question Generator*
4. *CSV + AI Insights App*



Capstone Deliverables

- *Streamlit app*

- *Clean Python code*
 - *Basic README*
 - *Demo walkthrough*
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Internship Outcomes

Skills Gained

- *Python programming*
- *Streamlit app development*
- *Basic AI / LLM usage*
- *API integration*
- *Real-world project exposure*

Certification

- *Internship Completion Certificate*
 - *Project-based evaluation*
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Evaluation Criteria

- *Code quality*
 - *Task completion*
 - *Understanding of concepts*
 - *Project presentation*
 - *Consistency & discipline*
-

Final Project Guidelines (Internship)

Constraints (Very Important)

-  **Static data only** (list / dictionary in Python)
 -  **Maximum 5 screens**
 -  **No database**
 -  **No authentication**
 -  **Streamlit-based UI**
 -  **Simple & readable code**
-

OPTION : Student Management App

(Recommended – easier to evaluate)

Project Overview

A simple **Student Management App** to **add, view, edit, and delete students** using static data.

Screen Breakdown (Max 5)

Screen 1: Dashboard (Home)

Purpose: Entry point of the app

UI Elements

- Title: *Student Management System*
- Buttons:

-  Add Student
-  View Students
-  Edit Student
-  Delete Student

Logic

- Use Streamlit sidebar or buttons to navigate
-

Screen 2: Add Student

Purpose: Add new student record

Fields

- Student ID
- Name
- Class
- Age
- Email

Actions

- Save button
- Validate empty fields

Static Data Example

```
students = [  
    {"id": 1, "name": "Amit", "class": "10", "age": 15, "email":  
    "amit@test.com"}  
]
```

Screen 3: View Students

Purpose: Display all students

Features

- Table view (`st.dataframe`)
 - Search by name (optional)
 - Display total count
-

Screen 4: Edit Student

Purpose: Modify existing record

Flow

- Select student by ID or name
 - Auto-fill form
 - Update values
 - Save changes
-

Screen 5: Delete Student

Purpose: Remove student

Flow

- Select student
 - Confirmation message
 - Delete from list
-

Skills Tested

- Python lists & dictionaries
- Streamlit forms

- State handling
 - CRUD concepts
 - UI navigation logic
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OPTION 2: Doctor Appointment Booking App

Project Overview

A basic **Appointment Booking App** using static doctor and appointment data.

Screen Breakdown (Max 5)

Screen 1: Home

Purpose: App introduction

UI

- App title
 - Buttons:
 - Book Appointment
 - View Appointments
-

Screen 2: Doctor List

Purpose: Show available doctors

Static Doctors

- Name
 - Specialization
 - Available time slots
-

Screen 3: Book Appointment

Purpose: Book appointment

Fields

- Patient Name
 - Doctor (dropdown)
 - Date
 - Time Slot
 - Reason
-

Screen 4: View Appointments

Purpose: Show booked appointments

Features

- Table view
 - Filter by doctor
-

Screen 5: Cancel Appointment

Purpose: Remove appointment

Flow

- Select appointment
- Confirm cancellation

Skills Tested

- Form handling
 - Dropdowns
 - Conditional UI
 - Static data manipulation
-

Suggested Folder Structure

```
final_project/  
| — app.py  
| — data.py  
| — utils.py  
| — README.md
```

Evaluation Checklist (For Interns)

Criteria	Marks
UI screens (≤ 5)	
Static data usage	
CRUD operations	

Code 
readability

Error handling 

App runs 
without crash

Recommendation

Choose Student Management App

- Easier logic
- Better CRUD practice
- Faster completion